

Aman Goel

 amangoel.ai |  aman@amangoel.ai |  Scholar |  ORCID |  GitHub |  LinkedIn

Research Focus

I spent a decade building automated provers — a competition-winning model checker (**1st, HWMCC 2020**) and the **first automatic proof of Lamport's Paxos** — and now apply that rigor to LLM safety, agentic AI, and post-training. At AWS, I shipped the neurosymbolic guardrails behind Amazon Bedrock's Automated Reasoning checks (**CAV 2026, 99.2% soundness**), and built zero-knowledge hallucination detection (**EMNLP 2025: 39% F1 gain, +9pp** on GPQA-diamond via cross-model test-time compute) and RL-guided red teaming (**NAACL 2025 oral: $\geq 95\%$ ASR** on GPT-4o/GPT-4 Turbo, $3\times$ fewer queries) deployed inside AWS. My thesis: **LLMs can't be trusted without proofs** — my work maps what formal verification can and cannot do about that. **17+** papers (CAV, EMNLP, NAACL, SOSP, ATC, FMCAD); prototype to production in months.

Experience

Amazon Web Services – Automated Reasoning Group

Seattle, WA

Senior Applied Scientist | Neurosymbolic AI

Nov 2021 – Present

- **Built Neurosymbolic Guardrails** | *Shipped as Automated Reasoning checks in Amazon Bedrock Guardrails · CAV 2026*
- Co-built the pipeline: LLM formalizes policies $\rightarrow$ SMT solver verifies claims $\rightarrow$ **99.2% soundness**; catches hallucinations pure-LLM approaches miss; generally available since Aug 2025, serving production traffic
- **Invented Zero-Knowledge Hallucination Detection** | *EMNLP 2025*
- Designed cross-model consistency framework requiring no external knowledge; **39% F1 improvement** over baselines; **9pp accuracy gain** on GPQA-diamond; rigorous evaluation on 5 benchmarks, 8 models
- **Built Automated Red Teaming System** | *NAACL 2025 Oral · Open-sourced*
- Engineered mutation-based fuzzer with RL-guided selection; **$\geq 95\%$ ASR** on GPT-4o/GPT-4 Turbo in **$3\times$ fewer queries**; deployed for internal red teaming
- **Owned Distributed Systems Verification** | *Aurora, DynamoDB, MemoryDB*
- Built systematic fuzzing + model checking tools; found critical design bugs missed by testing; drove adoption across 10+ AWS teams; reduced verification time from weeks to hours
- **Research leadership**: mentored 5+ scientists & engineers, supervised 4+ intern projects; PC member, **CAV 2025**; sustained independent formal-methods agenda alongside product work (CAV'25, ISoLA'24, FORTE'23, FMCAD'23)

University of Michigan

Ann Arbor, MI

PhD Researcher | Advisor: Karem Sakallah

Aug 2016 – Oct 2021

- **IC3PO: First Automatic Proof of Lamport's Paxos** | *FMCAD 2021*
- Invented symmetry-aware IC3 algorithm for infinite-state verification; enabled **Sift (ATC'22)**, **Bakery proof (FORTE'23)**
- **AVR: Competition-Winning Model Checker** | *1st HWMCC 2020 · medals at HWMCC 2024 & 2025*
- Engineered word-level abstraction-refinement; **7 gold medals**; benchmarked on 1000+ industrial designs

SRI International (intern) — **RL-guided quantifier instantiation** in the **Yices 2** SMT solver

Summer 2020

Cadence (research intern) — word-level verification engines for JasperGold; shipped to production

Summer 2019

Education

University of Michigan — Ph.D., Computer Science · advisor: Karem Sakallah

Aug 2016 – Oct 2021

Thesis: *From Finite to Infinite: Scalable Automatic Verification of Hardware Designs and Distributed Protocols*

IIT Madras — B.Tech EE + M.Tech Microelectronics · GPA 9.23/10, **Silver Medalist**

Jul 2011 – May 2016

Selected Publications

- 📄 **A Neurosymbolic Approach to Natural Language Formalization and Verification** – CAV 2026
C. An, S. Bayless, et al. (30 authors, alphabetical, incl. A. Goel) · to appear · deployed in Amazon Bedrock Guardrails
- 📄 **Zero-Knowledge LLM Hallucination Detection via Cross-Model Consistency** – EMNLP 2025
A. Goel, D. Schwartz, Y. Qi
- 📄 **TurboFuzzLLM: Mutation-based Fuzzing for Jailbreaking LLMs** – NAACL 2025 *Oral*
A. Goel, X. Wu, Z. Wang, D. Bepalov, Y. Qi
- 📄 **Sift: Refinement-guided Automation for Distributed Systems** – USENIX ATC 2022
H. Ma, H. Ahmad, A. Goel, et al.
- 📄 **Towards an Automatic Proof of Lamport's Paxos** – FMCAD 2021
A. Goel, K. Sakallah
- 📄 **I4: Incremental Inference of Inductive Invariants** – SOSP 2019
H. Ma, A. Goel, et al.

17+ publications at **CAV**, **EMNLP**, **NAACL**, **SOSP**, **USENIX ATC**, **FMCAD**, **TACAS**, **NFM**, **DATE**

Awards & Recognition

- 1st Place**, **Hardware Model Checking Competition 2020** – 7 Gold, 1 Silver, 1 Bronze (vs. industry tools)
- Medals**, **HWMCC 2024 & 2025** – gold + silver (2024) · silver, word-level arrays (2025)
- Rackham Predoctoral Fellowship 2020** – Top PhD research at Michigan · Simons Institute visitor (2021)
- Best Student Research Award** – CSE Graduate Honors Competition 2019
- Silver Medalist** – IIT Madras, Branch Rank 2/120 in Electrical Engineering
- National Award** for Empowerment of Persons with Disabilities – Govt. of India, 2013

Technical Skills

- Neurosymbolic & Formal:** SMT solvers (**Z3**, **Yices**), **Lean**, IC3/model checking, NL-to-formal-spec translation, proof automation
- LLM Safety & Post-Training:** hallucination detection, jailbreak fuzzing & red teaming, guardrails, fine-tuning & post-training, agentic AI, test-time compute (MCTS, cross-model consistency), LLM-as-judge evaluation
- Engineering:** Python, C++, C, Java | PyTorch | AWS (Bedrock, SageMaker) | large-scale evaluation pipelines

Open Source & Research Tools

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| 🔗 TurboFuzzLLM – LLM red-teaming fuzzer (NAACL 2025 oral); RL-guided mutation fuzzing | Python |
| 🔗 AVR – Hardware model checker; HWMCC 2020 winner; used by Cadence, academia | C++, Python |
| 🔗 IC3PO – Protocol verifier; first automatic proof of Lamport's Paxos (to our knowledge) | Python |
| 🔗 Contributor: Yices 2 (SMT solver), Yosys (synthesis) | C, C++ |

Service & Leadership

- Program Committee:** **CAV 2025** (top formal methods venue), **FMCAD 2022**
- Artifact Evaluation:** SOSP 2023/2021, CAV 2023/2020, OSDI 2021, MICRO 2021
- Mentorship:** Mentored 5+ scientists & engineers; supervised 4+ intern projects on formal verification and AI safety
- Press:** **Hacker News front page** (159 points, 68 comments) · **Michigan Engineering** features · **AWS News Blog**, Automated Reasoning checks GA
- Talks:** **FMCAD 2021 Paxos talk** (recorded) · Panelist, **ACL 2026 mentorship session** on AI tools & career growth